No Class Monday

Wednesday 3/4

Continuous Least Squares

→ given a function, approximate using a "simple" function

→ goal: assume f is continuous on $[a,b]$, find the n^{th} degree polynomial $P_n(x)$ st the distance

$$\int_a^b [f(x) - P_n(x)]^2 dx$$

for $f, g \in C[a,b]$, the inner product is defined as...

$$\langle f, g\rangle = \int_a^b f(x)g(x)dx$$

↳ recall: $\vec{a}, \vec{b} \in \mathbb{R}^2$ $\quad \vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2$

⋮ Properties of $\langle \cdot, \cdot \rangle$

i) $\langle f,f\rangle \geq 0$ & $\langle f,f\rangle = 0$ iff $f=0$

ii) $\langle f,g\rangle = \langle g,f\rangle$

iii) $\langle f+g, h\rangle = \langle f,h\rangle + \langle g,h\rangle$

Euclidean norm: $||f||^2 = \langle f,f\rangle = \int_a^b f^2(x)dx$

If $P_n(x)$ is the model...

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n = \sum_{i=0}^n a_ix^i$$

$0 \leq ||f - P_n||^2$

$= \langle f(x) - P_n(x), f(x) - P_n(x)\rangle$

$= \langle f(x), f(x)\rangle - 2\langle f, P_n\rangle + \langle P_n, P_n\rangle$

$= \langle f,f\rangle - 2\langle f, \sum_{i=0}^n a_ix^i\rangle + \langle \sum a_ix^i, \sum a_ix^i\rangle$

$= \int_a^b f^2(x)dx - 2\sum_{i=1}^n a_i \int_a^b f(x)x^i dx + \int_a^b \left(\sum_{i=1}^n a_ix^i\right)^2 dx$

ex)

Consider $\frac{\partial G}{\partial a_k} = 0$, $K = 0, 1, \dots, n$

$\frac{\partial G}{\partial a_k} = -2\langle f, x^k\rangle + 2\int_a^b \left(\sum_{i=0}^n a_ix^i\right)x^k dx = 0$

for each $k = 0, 1, \dots n$

} Brute Force Method

$$0 = -2\langle f, x^k\rangle + 2\langle P_n, x^k\rangle = 0$$

$\rightarrow \langle P_n, x^k\rangle = \langle f, x^k\rangle \quad k = 0, \ldots, n$

$\rightarrow \langle \sum_{i=0}^{n} a_i x^i, x^k\rangle = \langle f, x^k\rangle$

$\sum_{i=0}^{n} a_i \langle x^i, x^k\rangle = \langle f, x^k\rangle$

$$\vec{x} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} \quad \vec{b} = \begin{bmatrix} \langle f, x^0\rangle \\ \langle f, x^1\rangle \\ \vdots \\ \langle f, x^n\rangle \end{bmatrix}$$

$\langle x^i, x^k\rangle = \int_a^b x^{i+k}\,dx$

$$A = \begin{bmatrix} \langle x^0, x^0\rangle, & \langle x^1, x^0\rangle & \ldots & \langle x^n, x^0\rangle \\ \langle x^0, x^1\rangle & \langle x^1, x^1\rangle & & \vdots \\ \vdots & & \ddots & \vdots \\ \langle x^0, x^n\rangle, & \langle x^1, x^n\rangle & \ldots & \langle x^n, x^n\rangle \end{bmatrix}$$

Hilbert Matrix
* ill-conditioned *
will give issues

note: $P_n(x) = \sum_{i=0}^{n} a_i x^i$

$\rightarrow$ what if $P_n(x) = \sum_{i=0}^{n} a_i \varphi_i(x)$

where $\{\varphi_i\}_{i=0}^{n}$ is a basis for P_n (orthonormal)

$\rightarrow$ will see that orthogonality avoids ill-cond. matrix

Friday 3/6

Continuous Least Squares

ex | approx. $f(x) = x^3$ on $[0, 2]$ with a linear func: $P_1(x) = a_0 + a_1 x$

$$\begin{bmatrix} \langle x^0, x^0\rangle & \langle x^0, x^1\rangle \\ \langle x^1, x^0\rangle & \langle x^1, x^1\rangle \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \langle x^0, f\rangle \\ \langle x^1, f\rangle \end{bmatrix}$$

* try plotting in desmos first

$\int_0^2 dx = 2$

$$\begin{bmatrix} 2 & 2 \\ 2 & 8/3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 32/5 \end{bmatrix}$$

$$P_1(x) = -1.6 + 3.6x$$

↳ poorly conditioned.... let's improve

note: $\{x^i\}_{i=0}^n$ are <u>not</u> a good basis

→ might want to use $\{\varphi_i\}_{i=0}^n$ such that

$$P_n(x) = \sum_{i=0}^{n} a_i \varphi_i(x)$$

if $\langle \varphi_i, \varphi_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

$$\therefore \quad G = \langle f, f \rangle - 2\sum_{i=0}^{n} a_i \langle f, \varphi_i \rangle + \sum_{i=0}^{n} a_i^2$$

$$\frac{\partial G}{a_k} = -2\langle f, \varphi_k \rangle + 2a_k = 0$$

$$a_k = \langle f, \varphi_k \rangle$$

$\therefore$ our matrix → sparce

$$\begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} \langle f, x^0 \rangle \\ \langle f, x^1 \rangle \\ \vdots \\ \langle f, x^n \rangle \end{bmatrix}$$

$$P_n(x) = \sum_{i=0}^{n} \langle f, \varphi_i \rangle \varphi_i(x)$$

→ can get orthonormal using Graham Schmidt

ex) approx. $f(x) = x^3$ on $[0, 2]$ with

$\varphi_0 = 1 \quad \varphi_1 = x - 1 \quad \varphi_2 = x^2 - 2x + 2/3$ } orthogonal, not orthornormal

$\therefore$ off by constant

$$\begin{array}{c} \langle \varphi_0, \varphi_0 \rangle \\ \\ \end{array} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \langle \varphi_0, f \rangle \\ \langle \varphi_1, f \rangle \end{bmatrix}$$

$\langle \varphi_1, \varphi_1 \rangle$

$$a_0 = \frac{\langle \varphi_0, f \rangle}{\langle \varphi_0, \varphi_0 \rangle}, \quad a_1 = \frac{\langle \varphi_1, f \rangle}{\langle \varphi_1, \varphi_1 \rangle}$$